

Visal Gunawardena

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PROFILE

A motivated Mechanical Engineering undergraduate at the University of Moratuwa with a strong interest in mechanical systems, aerodynamics, and product development. Experienced in 3D CAD design, simulation, and hands-on fabrication. Recognized for leadership, adaptability, and teamwork in both academic and extracurricular settings. Eager to apply technical and creative skills in a R&D or industrial environment.

EDUCATION

University of Moratuwa	2022 - present
Bsc. Mechanical Engineering (Hons.)	
Kingswood College - Kandy	2019 - 2021
Passed G.C.E Advanced Level obtaining 3 A's	Z Score: 2.1107
Wesswood International College - Kandy	2008 - 2018
Passed G.C.E Ordinary Level obtaining 8 A's and 1 B.	

PROJECTS

Sliding Mesh Gearbox design for a Corn Shredder, (Semester 5 Project)

Designed and modeled a 3F+1R sliding-mesh gearbox for a corn-shredding machine, optimizing manufacturability, safety, and assembly; developing the solution using Solid Edge for local farming applications.

Production and Process Analysis of Perez Leather Artisans, (Semester 5 Project)

Analyzed leather goods manufacturing process to identify inefficiencies and resource gaps; used Tecnomatix Plant Simulation to optimize processes, reduce costs, minimize waste, and boost productivity and quality.

GridX Stem Challenge 2025

Designed and fabricated a CO_2 powered mini race car using Fusion 360 and ANSYS CFD. Optimized aerodynamic drag to improve speed; achieved Runner-up (Fastest Car Category) in GridX STEM Challenge 2025.

Skyward Challenge 2.0 Techno 2025

Designed and constructed a hand-launched glider; performed aerodynamic analysis to identify optimal flight conditions and structural parameters. Built the glider on site within 3 hours and achieved 4th place overall.

IMechE Railway challenge 2026, (Ongoing competition)

Participating in the IMechE Railway Challenge 2026, currently laying the groundwork and initial development for the project. Engaged in conceptual design, planning, and preparatory tasks as part of the ongoing development process.

Reverse Engineering of a Table Lamp, (Semester 4 Project)

Developed a SolidWorks prototype of a table lamp to analyze its manufacturing processes, performed material testing and stress analysis to evaluate performance and durability, and proposed design modifications to enhance functionality, aesthetics, and manufacturability.

Filerover: Automated files distributing robot, (Semester 2 Mechatronic Project)

Designed and built a fully automated file-distribution robot to streamline office document delivery, ensuring timely and accurate handoffs by monitoring personnel availability.

Antigravity table, (Semester 2 Manufacturing Project)

Designed and fabricated an antigravity table using woodworking and welding, combining multiple manufacturing techniques to achieve durability, aesthetics, and precision.

SKILLS

Technical Tools

- SolidWorks, SolidEdge, Fusion 360, AutoCAD, ANSYS, MATLAB, Arduino, Tecnomatix Plant Simulation

Programming

- Python, C++

Soft Skills

- Team leadership, problem-solving, communication, adaptability

SOCIETIES AND LEADERSHIP

Batch Representative of 22nd Batch , Department of Mechanical Engineering	2025 - Present
• Mechanized 24' – Organizing Lead • Sililara Handawa - Organizing Committee	
University Basketball Team , Member	2022 -Present
• Part of the winning team at Sri Lanka University Games (SLUG), 2025 • LTTL 5x5 Challenge Trophy – Organizing Committee • MyCola 3x3 Challenge Trophy - Organizing Committee	
IESL Student Chapter , University of Moratuwa	2024
• Exhibition Committee – Pillar Member • Compere	
Batch Representative (E group) , Engineering Faculty	2022
• Faculty Pirith Ceremony – Organizing Lead	

References

Dr. Lihil U. Subasinghe

Senior Lecturer

Department of Mechanical Engineering

University of Moratuwa, Sri Lanka

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